2022.12.12 汇报

A.5

6.(a) 令 $w=e^{S(z)}$，代入，

$$S''+S'^2+(\lambda-\alpha z-\beta z^4)=0$$

可 $S'^2-\beta z^4=0$

即 $S'=\pm\sqrt{\beta}z^2 \Rightarrow S_{(x)}=\pm\frac{\sqrt{\beta}}{3}z^3$

取 $S_{0(x)}=-\frac{\sqrt{\beta}}{3}z^3$，令 $S_{(x)}=-\frac{\sqrt{\beta}}{3}z^3+S_{1(x)}$

代入，

$\Rightarrow$ $S_1'=\pm\frac{\alpha}{2\sqrt{\beta}}$

$S_1=\pm\frac{\alpha}{2\sqrt{\beta}}z$

令 $S_{(x)}=-\frac{\sqrt{\beta}}{3}z^3\pm\frac{\alpha}{2\sqrt{\beta}}z+S_{2(x)}$

代入，$S_2'=-\frac{1}{2}$

$\Rightarrow$ $S_2=-\ln z$

即 P:

主项 $=Ae^{-\frac{\sqrt{\beta}}{3}z^3-\frac{\alpha}{2\sqrt{\beta}}z-\ln z}\cdot z^{-1}$

A.6

1、代入罗德里格，

(a) $P_{2n(0)}=\frac{1}{2^{2n}(2n)!}\frac{d^{2n}}{dx^{2n}}(x^2-1)^{2n}=\frac{(-1)^n(2n)!}{(n!)^2\cdot 2^{2n}}$

$P'_{2n+1(0)}=\frac{(-1)^n}{2^{2n+1}}\frac{(2n+1)!}{(n+1)!n!}\cdot(2n+1)$

(b) $P_n'(\pm1) = \begin{cases} \frac{1}{(n-1)!}, & n=2k+1,\ k=0,1,2,\cdots \\ \pm\frac{1}{(n-1)!}, & n=2k,\ k=1,2,\cdots \end{cases}$

$P_n''(\pm1) = \begin{cases} \frac{1}{(n-2)!}, & n=2(k+2),\ k=0,1,\cdots \\ \frac{\pm1}{(n-2)!}, & n=2k+1,\ k=0,1,\cdots \end{cases}$

2、

$\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}P_k(x)\right]+k(k+1)P_k(x)=0 \Rightarrow P_l(x)\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}P_k(x)\right]+k(k+1)P_k(x)P_l(x)=0$

同理, $P_k(x)\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}P_l(x)\right]+l(l+1)P_k(x)P_l(x)=0$

相减 $\Rightarrow \left[k(k+1)-l(l+1)\right]P_l(x)P_k(x) - 2xP_l(x)P_k'(x)+2xP_k(x)P_l'(x)+(1-x^2)\left[P_l(x)P_k''(x)-P_k(x)P_l''(x)\right]=0$

$\Rightarrow \frac{\frac{d}{dx}\left[(1-x^2)\left(P_k'(x)P_l(x)-P_l(x)P_k'(x)\right)\right]}{k(k+1)-l(l+1)} = -P_l(x)P_k(x)$

积分, $\int_x^1 P_k(x)P_l(x)dx = \frac{(1-x^2)\left[-P_k'(x)P_l(x)+P_l(x)P_k'(x)\right]}{k(k+1)-l(l+1)}\Big|_x^1 = \frac{(1-x^2)\left[P_k'(x)P_l(x)-P_l(x)P_k'(x)\right]}{k(k+1)-l(l+1)}$

得证。

3、

(a) $A_k = -\int_{-1}^{1}(1+x)^k \frac{1}{l(l+1)}\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}P_l(x)\right]dx$

$= \frac{-1}{l(l+1)}\left[0-\int_{-1}^{1}k(1+x)^{k-1}(1-x^2)\frac{d}{dx}P_l(x)\,dx\right]$

$= -\frac{k}{l(l+1)}\left[2kA_{k-1}-(k+1)A_k\right]$

$\Rightarrow A_k = \frac{2k^2}{k(l+1)-l(l+1)}A_{k-1} \quad \Gamma(k) \quad 2k^2\cdot(k-1)^2$

又 $A_k = \int_{-1}^{1}(1+x)^k P_l(x)\,dx = \int_{-1}^{1}x^k P_l(x)dx = 2^{l+1}\frac{(l!)^2}{(2l+1)!}$

$\Rightarrow A_k = \frac{2^{k+1}(k!)^2}{(2l+1)!\,[k(k+1)-l(l+1)]!}$, $k\geq l$; $A_k=0$, $k<l$

(5) $k=l$ 时,

$$\int_0^1 P_k(x)P_l(x)dx=\frac{1}{2}\int_{-1}^{1}P_kP_l dx=\frac{1}{2l+1}$$

$k\neq l$ 时, 由 2, 可知,

$$\int_0^1 P_k(x)P_l(x)dx=\frac{k!\,l!\left[\frac{\sin\frac{k\pi}{2}\cos\frac{l\pi}{2}}{\frac{k}{2}\left(\frac{k}{2}-1\right)\cdots\left(\frac{l}{2}+1\right)}-\frac{\sin\frac{l\pi}{2}\cos\frac{k\pi}{2}}{\frac{l}{2}\left(\frac{l}{2}-1\right)\cdots\left(\frac{k}{2}+1\right)}\right]}{2^{k+l-1}[k(k+1)-l(l+1)]}$$

(6) $\int_{-1}^{1}P_l(x)\ln(1-x)dx=\frac{1}{l(l-1)}\int_{-1}^{1}\ln(1-x)\frac{d}{dx}\left[(x^2-1)\frac{d}{dx}P_l(x)\right]dx$

$$=\frac{1}{l(l-1)}\left[(x^2-1)\ln(1-x)\frac{d}{dx}P_l(x)\Big|_{-1}^{1}-\int_{-1}^{1}(x+1)\frac{d}{dx}P_l(x)dx\right]$$

$$=\frac{-1}{l(l-1)}\left[2P_l(1)-\int_{-1}^{1}P_l(x)dx\right]$$

$$=-\frac{2}{l(l-1)}$$